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**Master of Computer
Applications Semester – III**

Data Mining

**Title:
Smart Campus : Student Experience & Engagement
Analytics**

Submitted By

Name : L Prajwal

Reg No: 2511022250099

Date: 19-08-2026

Submitted To:

Faculty Name: Nagasri p

**Department of Computer
Application Alliance University
Chandapura - Anekal Main Road,
Anekal Bengaluru – 562106**

1. Abstract

This report presents a complete data mining pipeline built on a real university Student Experience & Engagement Survey containing 145 responses across 18 raw fields and 6 academic departments (B.Tech, BCA, BSc, M.Tech, MCA, MSc). The project moves from raw, messy survey data through cleaning and feature engineering, unsupervised clustering, supervised classification, and finally into a deployed interactive dashboard, named Smart Campus Analytics, that a university administrator could use without writing any code.

Eleven facility and engagement dimensions — classrooms, faculty interaction, Wi-Fi, cafeteria, library, sports, clubs, technical events, cleanliness, safety, and transport — were extracted automatically from free-text survey column headers using keyword matching. These were combined into a single composite Happiness Index (0–100) for every student. K-Means clustering ($k = 3$) grouped students into Low, Moderate, and High satisfaction segments, while a Decision Tree classifier predicted overall satisfaction status with 93.1% test accuracy and identified club/extracurricular activities and cafeteria quality as the two strongest drivers of dissatisfaction. The final Gradio-based dashboard exposes department-level drill-downs and a live “what-if” student simulator, closing the loop between analysis and decision-making.

2. Introduction

Universities routinely collect student feedback through surveys, yet this feedback is often summarized only as raw averages in a spreadsheet and rarely mined for deeper, actionable patterns. The goal of this project was to treat a real Google Forms–style survey export as a data mining problem: clean an inconsistent, human-authored spreadsheet, engineer a set of comparable numeric features from long free-text question headers, discover natural groupings of students by satisfaction level, build a predictive model that explains what drives dissatisfaction, and present all of this through an interface that non-technical stakeholders (deans, student affairs staff) could actually use.

The finished system, Smart Campus Analytics, therefore sits at the intersection of three data mining tasks that are typically taught separately: exploratory data analysis, clustering (unsupervised learning), and classification (supervised learning) — unified behind one dashboard.

3. Objectives

- Clean and standardize a real-world, semi-structured survey export (mixed column types, missing values, duplicate rows, inconsistent headers).
- Automatically engineer 11 comparable satisfaction features from long-form survey question text, without hard-coding column positions.
- Construct a single composite metric (the Happiness Index) to summarize each student's overall experience on a 0–100 scale.
- Segment students into meaningful satisfaction groups using unsupervised clustering (K-Means).

- Train a supervised classifier to predict whether a student is “Satisfied” or “Not Satisfied,” and interpret which campus facilities matter most.
- Deploy the full analysis as an interactive, filterable dashboard with a live single-student simulator for administrators.

4. Dataset Description

The source data is an Excel export of a Google Forms survey, “University Student Experience & Engagement Survey (Responses).” After removing fully empty rows/columns and duplicate submissions, the working dataset contained 145 student responses and 18 columns.

Attribute	Description
Timestamp	Survey submission date/time
Program/Department	Student's degree program — B.Tech, BCA, BSc, M.Tech, MCA, MSc
Semester	Current semester of study
11 Satisfaction Questions	5-point Likert-scale ratings (1 = Very Poor – 5 = Excellent) covering classrooms, faculty, Wi-Fi, cafeteria, library, sports, clubs, technical events, cleanliness, safety, and transport
Participation Level	Self-reported activity/engagement in campus events and clubs
Overall Satisfaction	Overall satisfaction rating with university experience
Open-ended fields (2)	Free-text responses on improvement areas and suggestions

Table 1. Summary of raw dataset attributes.

5. Methodology

5.1 Data Cleaning

The raw export required standard cleaning before any analysis: rows and columns that were entirely empty were dropped, duplicate submissions were removed, and column headers were stripped of leading/trailing whitespace, since Google Forms headers frequently carry stray spaces that break exact-match lookups. This step reduced noise without discarding any genuine student responses, leaving the working set at 145 rows and 18 columns.

5.2 Automated Feature Extraction

Rather than hard-coding column indices — which would break if the survey were re-ordered or re-worded — the notebook maps each of the 11 target facility dimensions to a small set of keywords (for example, “wifi,” “wi-fi,” “wi fi,” and “internet” all resolve to the Wi-Fi feature). Each survey column is scanned against these keyword lists, matched columns are coerced to numeric type, and any missing values are imputed with the column median. This produced 11 clean, directly comparable satisfaction features:

Classrooms, Faculty, Wi-Fi, Cafeteria, Library, Sports, Clubs, Tech Events, Cleanliness, Safety, and Transport. The same keyword-matching approach was used to auto-detect the department/program column.

5.3 Composite Happiness Index

For every student, the 11 facility ratings were averaged into a single Average Rating (1–5 scale), then rescaled to a 0–100 Happiness Index using the transformation $(\text{Average Rating} - 1) / 4 \times 100$. Students were further bucketed into four qualitative categories — Very Happy (≥ 80), Happy (≥ 60), Neutral (≥ 40), and Needs Improvement (< 40) — to make the metric interpretable to non-technical stakeholders at a glance.

5.4 Exploratory Data Analysis

Across all 145 students, the overall average facility rating was 3.60 out of 5, corresponding to an average Happiness Index of 65.08 out of 100. A horizontal bar chart of mean rating per facility (colored on a continuous scale) was used to visually rank which services students rated best and worst, giving an immediate, ungrouped picture of institutional strengths and weaknesses before any modeling was applied.

5.5 Unsupervised Learning — K-Means Clustering

The 11 standardized facility features (via z-score scaling with StandardScaler) were passed into a K-Means model with $k = 3$ clusters ($\text{random_state} = 42$, $n_init = 10$). Clusters were then ranked and relabeled by their mean Happiness Index into Low Satisfaction, Moderate Satisfaction, and High Satisfaction segments, so the labels are meaningful rather than arbitrary cluster numbers.

Segment	Students	Share of Sample
Low Satisfaction	16	11.0%
Moderate Satisfaction	70	48.3%
High Satisfaction	59	40.7%

Table 2. K-Means cluster sizes ($k = 3$), ranked by mean Happiness Index.

Nearly half the student body falls into the Moderate Satisfaction segment, indicating that most students are neither strongly delighted nor strongly dissatisfied — a group that is often the most responsive to targeted, low-cost interventions.

5.6 Supervised Learning — Decision Tree Classification

A binary target, Satisfaction (“Satisfied” if Happiness Index ≥ 60 , else “Not Satisfied”), was constructed and predicted from the 11 facility features using a Decision Tree classifier ($\text{max_depth} = 4$, $\text{random_state} = 42$) on an 80/20 train–test split. The model achieved 93.1% accuracy on the held-out test set, showing that a small, interpretable set of facility ratings is highly predictive of a student's overall satisfaction status.

Feature importances extracted from the trained tree revealed a clear hierarchy of what drives satisfaction outcomes:

Rank	Feature	Importance
1	Clubs / Extracurricular Activities	0.485
2	Cafeteria	0.179
3	Safety	0.096
4	Library	0.087
5	Transport	0.077
6	Classrooms	0.035
7	Faculty	0.028
8	Cleanliness	0.011
9–11	Wi-Fi, Sports, Tech Events	0.000 (unused by the tree)

Table 3. Decision Tree feature importances.

Clubs and extracurricular engagement alone account for roughly 49% of the model's decision-making weight, nearly three times the next-highest feature (cafeteria, ~18%). This strongly suggests that student clubs and social/extracurricular life are the single biggest lever for improving perceived campus satisfaction, ahead of academic or infrastructure factors such as classrooms or faculty interaction.

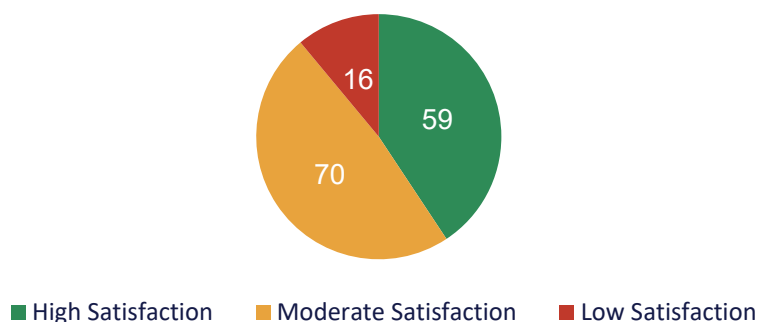
5.7 Interactive Dashboard (Gradio)

All analysis was wrapped into a two-tab Gradio web application:

- Live Dashboard — lets an administrator filter by department (or view “All”) and instantly see student count, average rating, Happiness Index, a facility satisfaction bar chart, a happiness-category pie chart, a department-wise happiness comparison, and a ranked list of the top concern areas.
- Student Analysis — a what-if simulator where an administrator enters a hypothetical student's 11 facility ratings via sliders and instantly receives a computed Happiness Index, satisfaction status, and a rule-based recommendation flagging any facility rated ≤ 2 as needing attention.

The dashboard was deployed with a public Gradio share link for demonstration purposes, making the model outputs accessible to non-technical users through sliders and dropdowns rather than code.

STUDENT SATISFACTION GROUPS (K-MEANS)



11 STUDENT EXPERIENCE FEATURES

1. Classrooms

2. Faculty Interaction

3. Wi-Fi

4. Cafeteria

5. Library

6. Sports

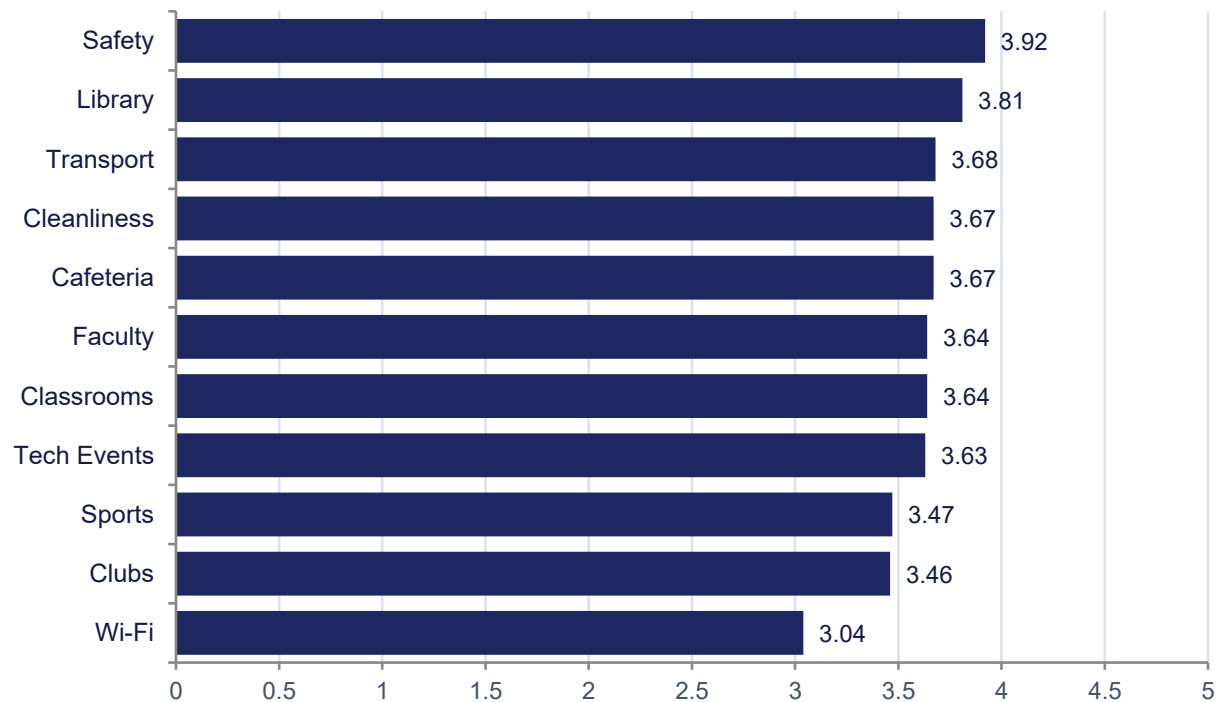
7. Clubs

8. Technical Events

9. Cleanliness

10. Safety

11. Transportation



6. Results Summary

Metric	Value
Total student responses (post-cleaning)	145
Facility dimensions analyzed	11
Overall average facility rating	3.60 / 5
Overall average Happiness Index	65.08 / 100
K-Means clusters	3 (Low / Moderate / High Satisfaction)
Largest satisfaction segment	Moderate Satisfaction (70 students, 48.3%)
Decision Tree test accuracy	93.1%
Top driver of dissatisfaction	Clubs / extracurricular activities (48.5% importance)
Second driver of dissatisfaction	Cafeteria quality (17.9% importance)

Table 4. Key results at a glance.

7. Discussion

The results converge on a consistent story from three independent angles — descriptive statistics, clustering, and classification. The raw EDA shows an overall satisfaction level that is respectable but not excellent (3.60/5). Clustering reveals this average masks real heterogeneity: while 40.7% of students are highly satisfied, 11% are clearly underserved and the largest group (48.3%) sits in an ambiguous middle zone. Most importantly, the classifier's feature importances give administrators a prioritized, evidence-based action list rather than a vague mandate to “improve everything.”

The dominance of the Clubs feature is a particularly actionable finding: it implies that satisfaction is driven less by core academic infrastructure (classrooms, faculty) and more by the richness of student social and extracurricular life. This is consistent with broader higher-education research suggesting that a sense of community and belonging is often a stronger predictor of student well-being than facility quality alone. Cafeteria quality emerging as the second-strongest driver similarly points to daily-life quality-of-service factors rather than purely academic ones.

A limitation worth noting is sample size: with 145 respondents split across 6 departments, department-level breakdowns in the dashboard (used for the department-wise happiness chart) are based on smaller sub-samples and should be interpreted with appropriate caution. The Decision Tree's shallow depth (max_depth = 4) was a deliberate trade-off favoring interpretability over marginal accuracy gains, which is appropriate given the goal of producing actionable, explainable insights for non-technical stakeholders.

8. Conclusion

This project demonstrates an end-to-end data mining workflow applied to a genuine institutional problem: turning an unstructured student feedback survey into a decision-support tool. By combining automated feature engineering, a composite satisfaction index, unsupervised clustering, and an interpretable supervised classifier, the analysis identifies not just how satisfied students are, but why — and delivers those insights through a live, filterable dashboard rather than a static report. The clearest actionable takeaway is that investment in student clubs, extracurricular programming, and cafeteria services would likely yield the largest measurable improvement in overall student satisfaction, ahead of classroom, Wi-Fi, or sports infrastructure spending.

9. Recommendations for University Administration

Based on the combined evidence from EDA, clustering, and the Decision Tree's feature-importance ranking, the following actions are recommended in priority order:

Priority	Area	Recommended Action
1	Clubs & Extracurriculars	Increase funding and staff support for student clubs; launch new interest-based clubs; publicize activity calendars to raise participation, since this is the single strongest driver of dissatisfaction.
2	Cafeteria	Audit food quality, hygiene, pricing, and menu variety; run a satisfaction pulse-survey specifically on dining services each semester.
3	Safety & Security	Expand CCTV coverage, lighting, and campus patrols, particularly in the areas flagged in the open-ended survey responses.
4	Library	Extend operating hours and expand digital resource subscriptions, especially for the Moderate Satisfaction segment identified by clustering.
5	Transport	Review bus routes, frequency, and communication of schedule changes to students.

Table 6. Prioritized, evidence-based recommendations.

Because the Moderate Satisfaction cluster is both the largest (48.3% of respondents) and, by definition, not yet strongly dissatisfied, it represents the highest-leverage target for these interventions: comparatively small improvements to clubs and cafeteria services are likely to shift a large share of this group into the High Satisfaction segment.

10. Project Workflow Overview

The end-to-end pipeline implemented in the accompanying notebook followed six sequential stages, each building on the output of the previous one:

- Stage 1 — Ingestion: Load the raw Excel survey export with pandas/openpyxl.
- Stage 2 — Cleaning: Drop empty rows/columns, remove duplicates, strip whitespace from headers.
- Stage 3 — Feature Engineering: Keyword-match survey questions to 11 standardized facility features and auto-detect the department column; compute the composite Happiness Index and category labels.
- Stage 4 — Unsupervised Modeling: Scale features and apply K-Means ($k = 3$) to segment students into satisfaction tiers.
- Stage 5 — Supervised Modeling: Train and evaluate a Decision Tree classifier; extract feature importances to explain the drivers behind satisfaction.
- Stage 6 — Deployment: Wrap all analysis in a two-tab Gradio dashboard (department drill-down view and a live student what-if simulator) for non-technical end users.

11. Future Work

- Collect a larger, more balanced sample across all departments and semesters to strengthen department-level and cohort-level conclusions.
- Apply text mining (e.g., topic modeling or sentiment analysis) to the two open-ended free-text improvement questions, which were collected but not yet analyzed.
- Experiment with additional classifiers (Random Forest, Gradient Boosting) and cross-validation to validate the robustness of the 93.1% accuracy and feature-importance ranking.
- Run the survey longitudinally each semester to track whether interventions (e.g., new clubs, cafeteria upgrades) measurably shift the Happiness Index over time.
- Deploy the Gradio dashboard on permanent hosting (e.g., Hugging Face Spaces) rather than a temporary share link for ongoing institutional use.

12. Tools & Technologies Used

Category	Tools
Language / Environment	Python 3, Jupyter/Colab Notebook
Data Handling	pandas, numpy
Visualization	Plotly Express
Machine Learning	scikit-learn (StandardScaler, KMeans, DecisionTreeClassifier, train_test_split, accuracy_score)
Deployment / UI	Gradio (Blocks, Tabs, Dropdown, Slider, Plot components)
Data Source	Excel export (openpyxl) of a Google Forms survey

Table 7. Technology stack.